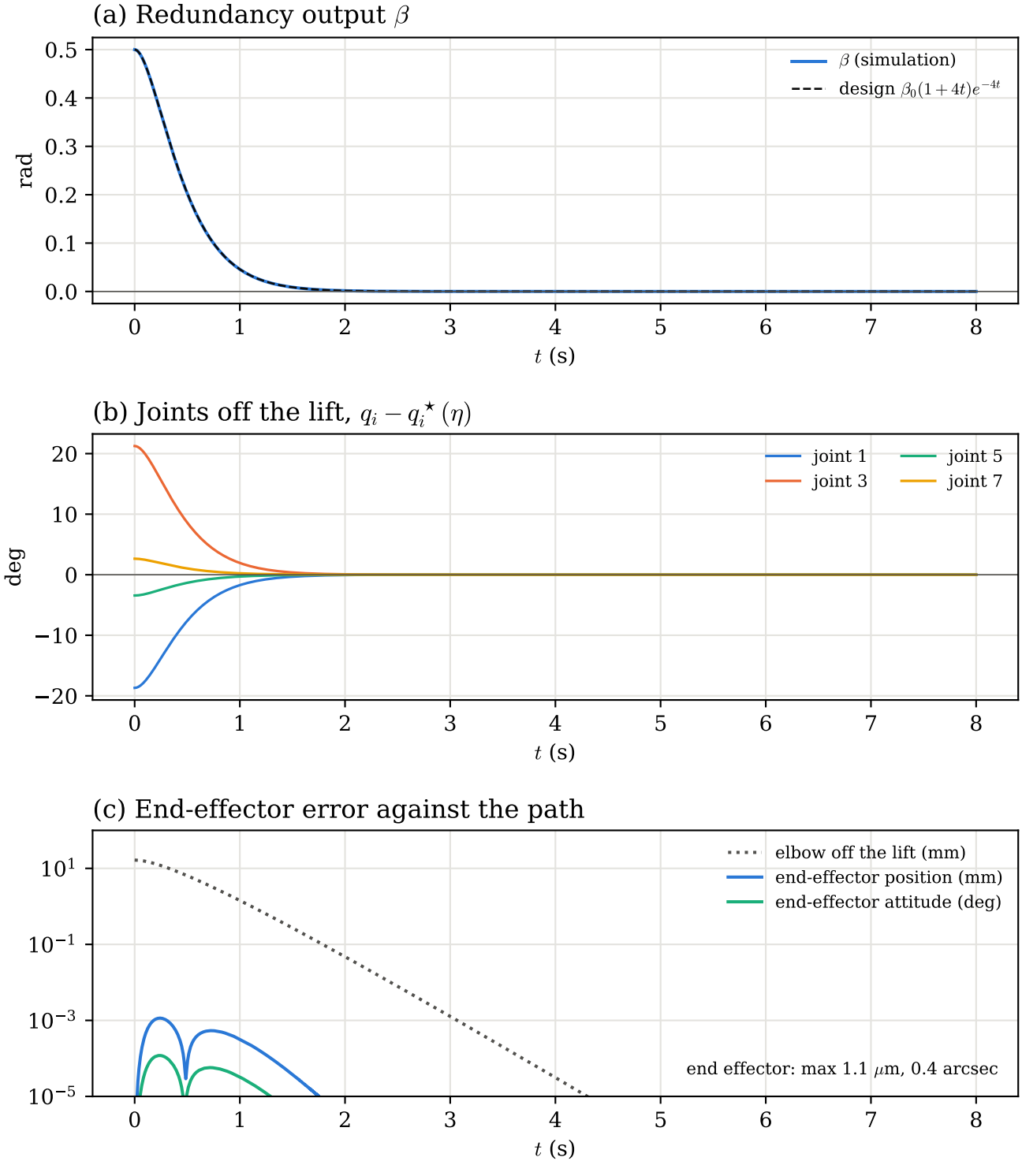

(a) Redundancy output β
rad
0.5
0.4
0.3
0.2
0.1
0.0
β (simulation)
design $\beta_0(1+4t)e^{-4t}$
0
1
2
3
4
5
6
7
8
t (s)
(b) Joints off the lift, $q_i - q_i^\star$ (η)
deg
20
10
0
−10
−20
joint 1
joint 3
joint 5
joint 7
0
1
2
3
4
5
6
7
8
t (s)
(c) End-effector error against the path
10^{1}
10^{-1}
10^{-3}
10^{-5}
elbow off the lift (mm)
end-effector position (mm)
end-effector attitude (deg)
end effector: max 1.1 μm, 0.4 arcsec
0
1
2
3
4
5
6
7
8
t (s)